

Alan Luu

Software Engineer – Backend & Data Platform

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EXPERIENCE

Azure CXP Software Engineer (Data & Backend)

Apr 2024 – Dec 2025

Microsoft

Redmond, WA

- Refactored legacy Scala ETL logic into PySpark in Azure Synapse, rewriting 20+ data pipelines that processed billions of Azure usage and billing records for Fortune 500 clients and Microsoft internal teams.
- Reduced runtime of 3 production Apache Spark pipelines by 75% (24h → 5–6h) by resolving stage retries, scaling worker capacity, improving partition pruning, and optimizing joins, accelerating customer reporting.
- Built end-to-end ETL data pipelines with Cosmos DB, Cosmos SCOPE, and Kusto/KQL to ingest, validate, and surface 80+ Azure telemetry metrics in a stakeholder-facing dashboard for usage and cloud performance reporting.
- Owned and enhanced 150+ Playwright E2E tests in TypeScript, expanding coverage across Product, Consumption, and Quality domains while stabilizing flaky tests and validating Azure dashboard features amid UI changes.

Xbox Software Engineer (Systems & Reliability)

Mar 2022 – Sep 2023

Microsoft

Redmond, WA

- Developed and maintained 6 Xbox reliability test suites in C# and C++, covering 20–60 automated tests per suite across hardware, firmware, OS, networking, and thermal validation for 9 Xbox console programs.
- Stabilized long-running Xbox reliability tests by resolving memory leaks, port exhaustion, cross-stack failures, and test infrastructure issues, unblocking technicians and engineers across 7 global hardware labs.
- Improved code quality and security across 4 production repositories by resolving 120+ CodeQL issues in C# and C++, addressing vulnerabilities and maintainability risks.

Xbox Cortana Software Engineer Intern (Full Stack)

Feb 2016 – Aug 2016

Microsoft

Redmond, WA

- Developed Cosmos SCOPE analytics scripts to aggregate Cortana speech usage data, helping engineering teams identify high-impact voice features for Cortana's release on Xbox One.
- Built OnePulse metric tiles and an ASP.NET dashboard in C# and JavaScript to visualize Cortana usage trends, improving access to voice feature usage and speech interaction insights.
- Supported Cortana validation by collecting speech feedback and reporting findings to improve release readiness.

PROJECTS

RetailLens | [GitHub](#) | [Live](#) | AWS, S3, Glue, PySpark, Athena, DynamoDB, FastAPI, React

June 2026

- Built a full-stack AWS retail analytics platform that ingests raw CSV data into S3, catalogs it with Glue, transforms it into partitioned Parquet using Glue PySpark, computes aggregate metrics with Athena, and serves precomputed analytics from DynamoDB through FastAPI and React.

Job Processing API | [GitHub](#) | [Live](#) | Python, FastAPI, SQLAlchemy, pytest, Docker

May 2026

- Built a job-processing backend for creating, claiming, and processing jobs through guarded state transitions, with SQLite persistence, an in-memory priority queue, pytest coverage, and Docker support.

EDUCATION

Bachelor of Science in Computer Science

Sep 2018 – Aug 2020

University of Washington

Bothell, WA

Associate in Arts and Science

Jan 2014 – Mar 2018

Bellevue College

Bellevue, WA

Apprenticeship in Data Analyst

Jun 2015 – Aug 2016

Year Up

Redmond, WA

TECHNICAL SKILLS

Languages: Python, SQL, C#, Java, JavaScript/TypeScript, C++, Scala

Cloud & Data: Azure, Azure Synapse, Azure Data Factory, Apache Spark/PySpark, ETL/ELT, Cosmos DB, Cosmos SCOPE, Kusto/KQL, Azure Blob Storage, SQL Server, PostgreSQL, MySQL, AWS, S3, Glue, Athena, DynamoDB, EC2

Frameworks & Libraries: FastAPI, SQLAlchemy, Pydantic, ASP.NET, React, Node.js, Next.js, Spring Boot

Developer Tools & Platforms: Azure DevOps (CI/CD), Git, pytest, Playwright, CodeQL, Docker, Linux, VS Code